Extended reanalysis of anti-amyloid monoclonal antibody trials

Individual drugs, trial termination, biological target class, absolute harms, MID compatibility, and exploratory moderator analyses

Purpose

This package investigates the additional empirical criticisms raised in Snyder et al. (2026) using the Cochrane CD016297 analysis export. It extends the prior plaque-clearance reanalysis without converting the evidence synthesis into a clinical-practice guideline or a single net-benefit judgment.

Headline

The data support treatment-specific reporting and absolute safety estimates. They do not establish statistically significant differences among antibodies, validate a single clinical-importance threshold, or prove that amyloid removal mediates individual benefit.

Prepared 14 August 2026. Aggregate-data meta-analysis; not individual medical guidance.

1. Executive summary

Question	Result	Interpretation
Do individual drugs differ?	Lecanemab and donanemab individually favored treatment on ADAS-Cog and CDR-SB.	Formal interaction tests were nonsignificant (ADAS-Cog P=0.433; CDR-SB P=0.395) and underpowered.
Do futility trials dilute efficacy?	ADAS-Cog SMD was -0.111 class-wide, -0.109 after excluding futility trials, and -0.123 after excluding all early terminations.	Little evidence that termination status alone explains the class estimate.
What are the absolute ARIA-E effects?	Excess per 1,000: aducanumab 326, donanemab 220, lecanemab 109.	Absolute risk differs materially by agent and is more useful clinically than a class RR alone.
Do effects reach cited MIDs?	At 18 months, the upper 95% CI remained below the cited 2-point ADAS-Cog and 1-point CDR-SB MCI benchmarks.	This is threshold compatibility, not proof of no patient value; thresholds were derived over shorter horizons.
Does ARIA-related unblinding explain efficacy?	ARIA-E risk-difference moderation was not detected for ADAS-Cog, MMSE, or CDR-SB.	The proxy is indirect and underpowered; absence of association does not eliminate functional unblinding.
Does target class explain effects?	Patterns differed descriptively across target classes.	Classes remain internally heterogeneous and strongly confounded with agent and trial era.

The most defensible conclusion is treatment-specific: modern plaque-clearing antibodies produce small average clinical effects with substantial, agent-dependent imaging toxicity. The expanded analyses weaken simple class-wide interpretations but do not justify a causal or universal benefit-risk claim.

2. Methods

The supplied Cochrane row-level data were annotated by antibody, biological target class, biomarker requirement, publication year, and trial-termination status. Termination coding followed the Cochrane review, which reports six studies terminated for futility, two for symptomatic ARIA-E, and one without a reported reason. The annotation reconciled exactly to those totals.

Statistical approach

Random-effects inverse-variance meta-analysis used REML heterogeneity estimation and Hartung-Knapp inference for three or more studies; Wald inference was used for one or two studies. A DerSimonian-Laird fallback was used only when REML failed to converge in a sparse subgroup. Risk ratios were analyzed on the log scale. Raw mean differences were reconstructed from arm means, standard deviations, and sample sizes where available.

New analyses

Analysis	Operationalization
Individual antibody	Separate pooled estimates for seven antibodies; omnibus treatment-by-antibody moderator tests when residual degrees of freedom permitted.
Trial termination	Class pool compared with exclusion of six futility-terminated trials and exclusion of all nine early-terminated trials.
Target class	Soluble monomer, soluble aggregate-preferring, aggregated/fibrillar, and pyroglutamate plaque categories.
Absolute safety	Risk difference, excess events per 1,000, and NNH/NNTB when the 95% CI did not cross zero.
MID compatibility	CIs compared with ADAS-Cog 2/3/4-point and CDR-SB 1/2-point benchmarks cited by Cochrane; no threshold newly endorsed.
Functional unblinding proxy	Trial-level efficacy regressed on excess ARIA-E risk per 10 percentage points.
Influence and moderation	Leave-one-antibody-out analyses and sparse one-moderator models for biomarker confirmation, futility, publication year, and ARIA-E risk.

Multiplicity

All new moderator, interaction, target-class, and leave-one-out analyses are exploratory. P values were not adjusted for multiplicity and should be read as descriptive measures of compatibility, not confirmatory tests.

3. Individual-antibody efficacy

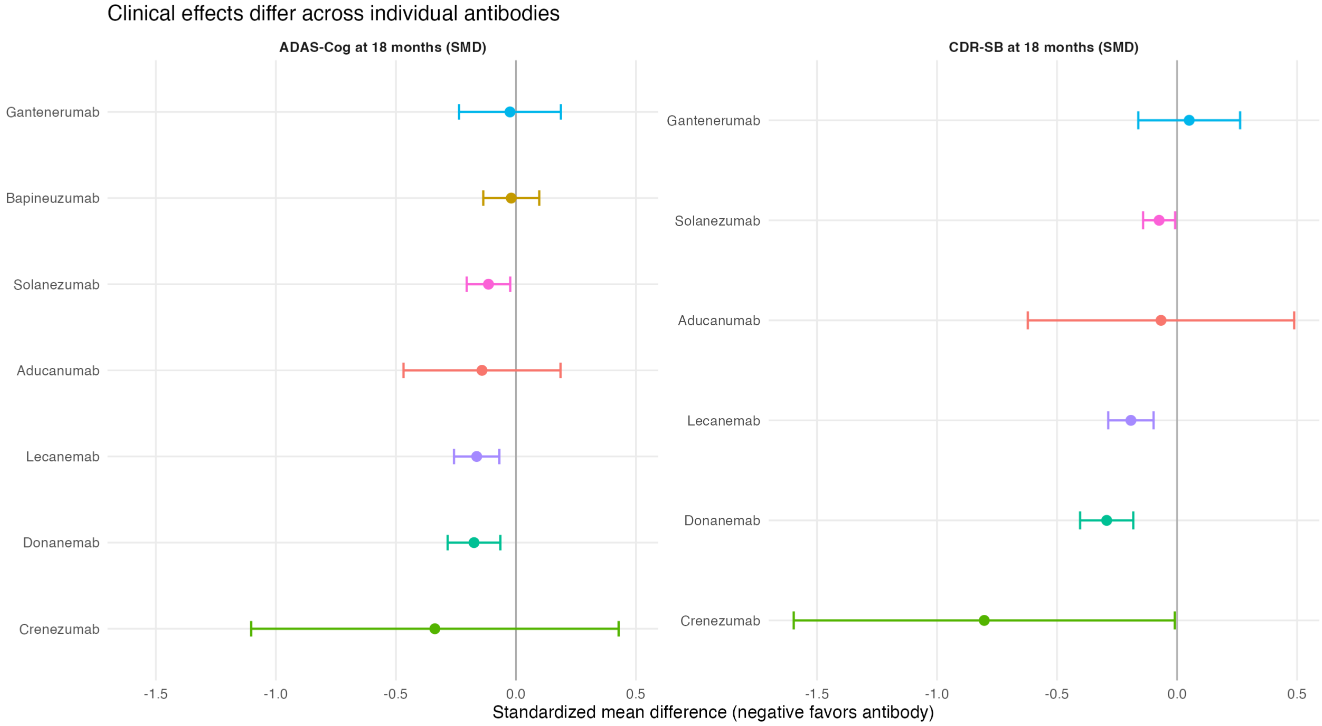

Figure 1. Individual-antibody standardized effects. Single-trial estimates retain their trial-level normal confidence intervals; multi-trial antibody pools use the prespecified random-effects method.

Agent	ADAS-Cog SMD (95% CI)	CDR-SB SMD (95% CI)	ADAS-Cog raw MD	CDR-SB raw MD
Aducanumab	-0.141 (-0.468, 0.186)	-0.067 (-0.622, 0.488)	-1.276 (-2.877, 0.325)	-0.037 (-1.295, 1.220)
Bapineuzumab	-0.020 (-0.136, 0.097)	NA	-0.125 (-1.149, 0.900)	NA
Crenezumab	-0.338 (-1.103, 0.428)	-0.803 (-1.597, -0.009)	-1.740 (-5.517, 2.037)	-1.300 (-2.485, -0.115)
Donanemab	-0.175 (-0.284, -0.065)	-0.293 (-0.404, -0.182)	-1.350 (-2.196, -0.504)	-0.700 (-0.964, -0.436)
Gantenerumab	-0.025 (-0.237, 0.187)	0.051 (-0.161, 0.263)	-0.230 (-2.195, 1.735)	0.130 (-0.413, 0.673)
Lecanemab	-0.164 (-0.258, -0.069)	-0.192 (-0.287, -0.098)	-1.440 (-2.270, -0.610)	-0.450 (-0.670, -0.230)
Solanezumab	-0.114 (-0.205, -0.024)	-0.075 (-0.141, -0.008)	-1.515 (-2.596, -0.434)	-0.248 (-0.468, -0.027)

Lecanemab and donanemab each favored treatment on both central endpoints. Aducanumab was heterogeneous and imprecise, especially for CDR-SB. Nevertheless, formal omnibus interaction tests did not establish between-antibody differences: P=0.433 for ADAS-Cog and P=0.395 for CDR-SB. With seven drugs but only 13 and nine contrasts, respectively, these tests have little power and cannot be treated as evidence of equivalence.

4. Trial-termination sensitivity analyses

Endpoint	Scenario	k	Estimate (95% CI)	P	I2
ADAS-Cog scale at 18 months	Cochrane class pool	13	-0.111 (-0.160, -0.062)	0.0003	10.1
ADAS-Cog scale at 18 months	Excl. futility-terminated trials	9	-0.109 (-0.172, -0.045)	0.0042	18.2
ADAS-Cog scale at 18 months	Excl. all early-terminated trials	6	-0.123 (-0.184, -0.062)	0.0036	9.0
CDR-SB scale at 18 months	Cochrane class pool	9	-0.124 (-0.243, -0.004)	0.0442	67.0
CDR-SB scale at 18 months	Excl. futility-terminated trials	5	-0.147 (-0.314, 0.019)	0.0695	72.5
CDR-SB scale at 18 months	Excl. all early-terminated trials	4	-0.155 (-0.326, 0.017)	0.0642	77.5
ADCS-ADL-MCI score at 18 months	Cochrane class pool	4	0.225 (0.116, 0.335)	0.0073	0.0
ADCS-ADL-MCI score at 18 months	Excl. futility-terminated trials	2	0.248 (0.149, 0.347)	0.0000	0.0
ADCS-ADL-MCI score at 18 months	Excl. all early-terminated trials	1	0.247 (0.148, 0.346)	0.0000	NA
Any ARIA E at 18 months	Cochrane class pool	11	10.022 (6.529, 15.383)	0.0000	25.2
Any ARIA E at 18 months	Excl. futility-terminated trials	8	8.762 (3.970, 19.337)	0.0003	53.7
Any ARIA E at 18 months	Excl. all early-terminated trials	6	9.477 (2.104, 42.696)	0.0121	83.7
Any ARIA H at 18 months	Cochrane class pool	3	1.648 (0.472, 5.750)	0.2278	90.0
Any ARIA H at 18 months	Excl. futility-terminated trials	2	2.138 (1.784, 2.563)	0.0000	25.6
Any ARIA H at 18 months	Excl. all early-terminated trials	2	2.138 (1.784, 2.563)	0.0000	25.6
Study discontinuation at 18 months	Cochrane class pool	11	1.121 (1.021, 1.231)	0.0218	49.6
Study discontinuation at 18 months	Excl. futility-terminated trials	8	1.095 (0.975, 1.229)	0.1064	57.7
Study discontinuation at 18 months	Excl. all early-terminated trials	6	1.124 (0.956, 1.320)	0.1221	58.9

Excluding futility-terminated trials did not strengthen ADAS-Cog and only modestly strengthened CDR-SB. Removing all early-terminated studies reduced the number of ADCS-ADL-MCI contrasts to one. For ARIA-H, removing futility trials changed the highly heterogeneous class estimate (RR 1.65; I2=90%) to the lecanemab-donanemab pool (RR 2.14; I2=26%), illustrating how class composition can strongly affect safety heterogeneity.

Interpretation

The response's concern is better supported by biological activity and individual drug than by termination status alone. Early termination is also a post-randomization program characteristic and may be related to efficacy or toxicity; exclusion can introduce selection bias.

5. Absolute safety effects

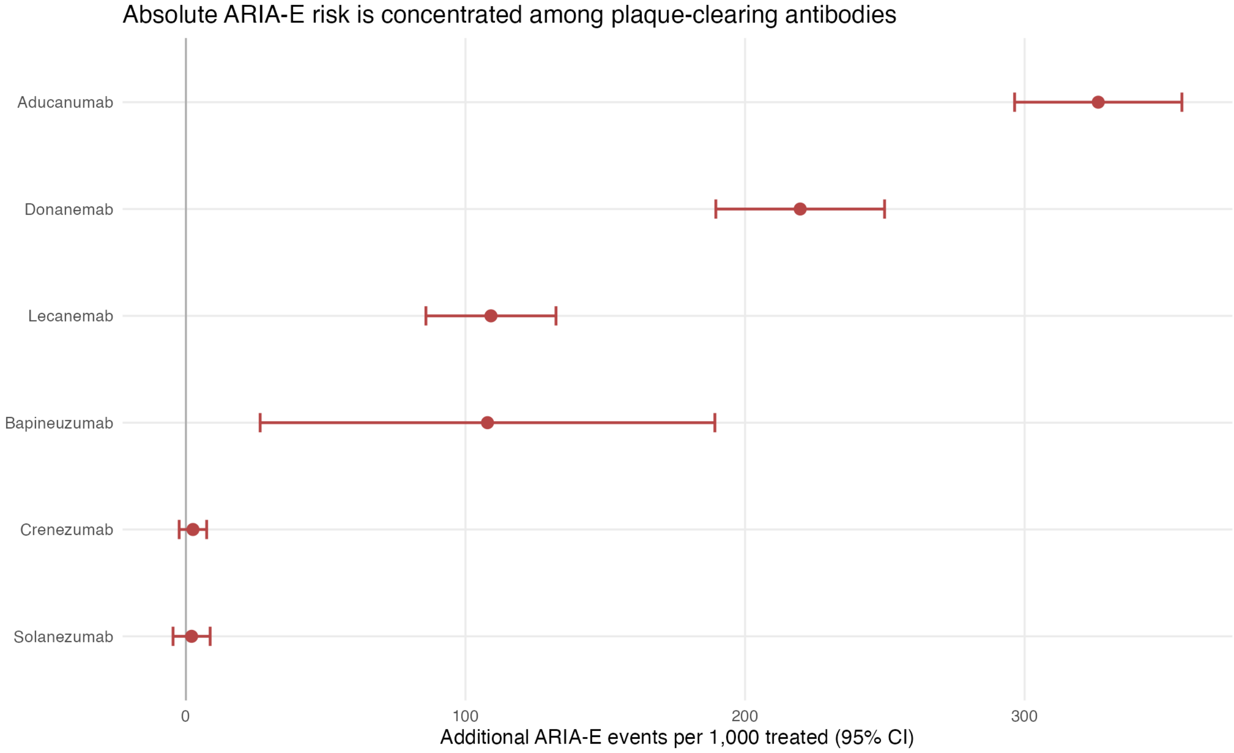

Figure 2. Agent-specific risk differences for any ARIA-E at approximately 18 months.

Outcome	Agent	Excess/1,000 (95% CI)	NNH or NNTB
Any ARIA E at 18 months	Aducanumab	326.3 (296.4, 356.3)	NNH 3.1
Any ARIA E at 18 months	Donanemab	219.7 (189.6, 249.9)	NNH 4.6
Any ARIA E at 18 months	Lecanemab	109.1 (85.9, 132.4)	NNH 9.2
Any ARIA H at 18 months	Donanemab	178.0 (139.5, 216.6)	NNH 5.6
Any ARIA H at 18 months	Lecanemab	82.3 (51.3, 113.3)	NNH 12.1
Study discontinuation at 18 months	Aducanumab	28.3 (5.3, 51.3)	NNH 35.4
Study discontinuation at 18 months	Donanemab	71.1 (31.5, 110.8)	NNH 14.1
Study discontinuation at 18 months	Lecanemab	32.1 (-2.8, 67.0)	Indeterminate
SAE at 18 months	Aducanumab	-5.8 (-34.7, 23.2)	Indeterminate
SAE at 18 months	Donanemab	15.6 (-19.5, 50.7)	Indeterminate
SAE at 18 months	Lecanemab	27.7 (-3.0, 58.4)	Indeterminate
Mortality at 18 months	Donanemab	7.3 (-4.2, 18.8)	Indeterminate
Mortality at 18 months	Lecanemab	-1.1 (-9.0, 6.7)	Indeterminate

ARIA-E excess risk was 326 per 1,000 for aducanumab (NNH 3.1), 220 per 1,000 for donanemab (NNH 4.6), and 109 per 1,000 for lecanemab (NNH 9.2). For ARIA-H, the excess was 178 per 1,000 for donanemab (NNH 5.6) and 82 per 1,000 for lecanemab (NNH 12.1). Discontinuation risk was clearly increased for donanemab and aducanumab; the lecanemab CI crossed zero.

These are trial-contrast averages, not personalized risks. APOE genotype, baseline MRI findings, anticoagulation, dose exposure, monitoring, and event definitions cannot be incorporated from the Cochrane aggregate export.

6. Treatment-specific benefit and harm

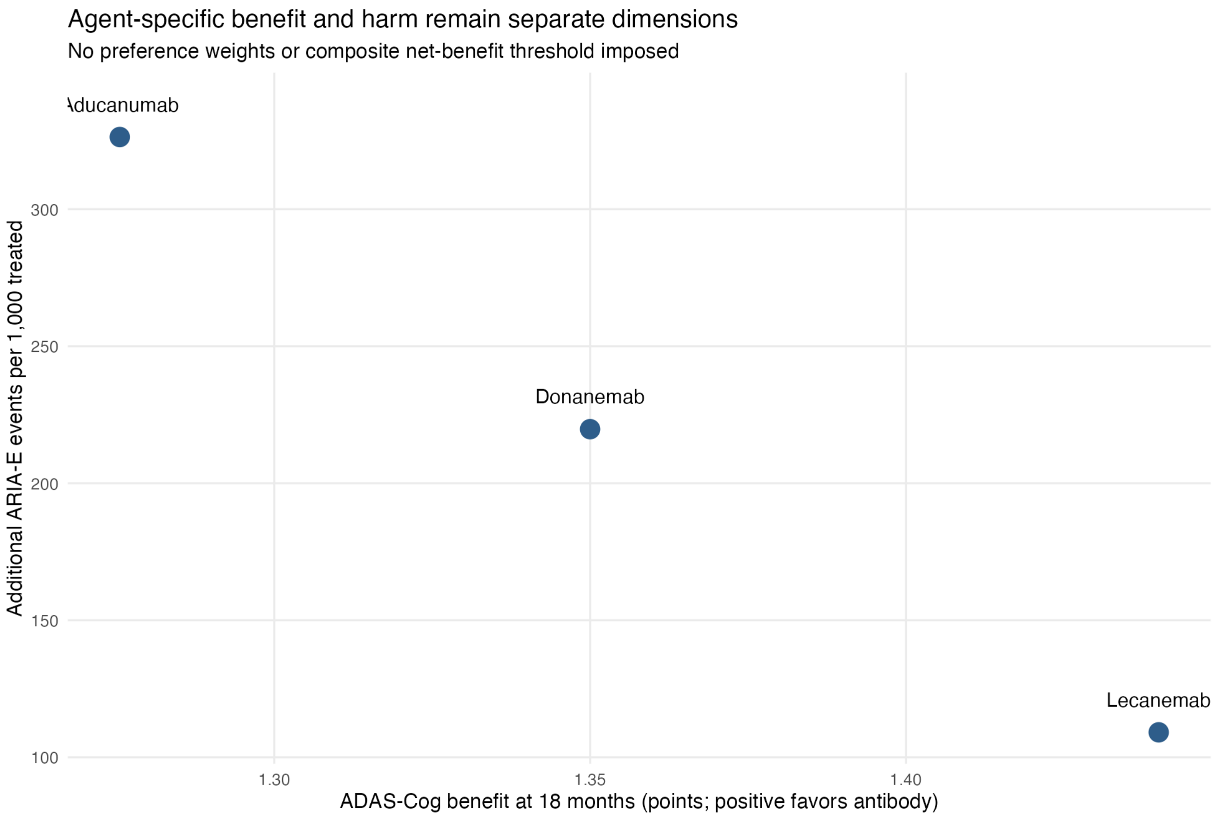

Figure 3. ADAS-Cog raw mean difference and ARIA-E risk difference are displayed as separate dimensions. The figure deliberately imposes no utility weights.

Agent	ADAS-Cog MD	CDR-SB MD	ARIA-E excess/1,000	Discontinuation excess/1,000
Aducanumab	-1.28	-0.04	326.3	28.3
Lecanemab	-1.44	-0.45	109.1	32.1
Donanemab	-1.35	-0.70	219.7	71.1

Lecanemab and donanemab had similar reconstructed ADAS-Cog differences (-1.44 and -1.35 points), while their CDR-SB differences were -0.45 and -0.70 points. Donanemab had larger absolute ARIA-E and discontinuation differences than lecanemab in these trials. Cross-trial comparisons are not randomized head-to-head comparisons and may reflect population, monitoring, estimand, and ascertainment differences.

No single benefit-risk ranking is calculated. Such a ranking would require patient preference weights, event severity and reversibility, treatment burden, cost, and guideline-level deliberation.

7. MID and MCID compatibility

The response correctly distinguishes statistical evidence from judgments about patient importance. To make that distinction explicit, raw-effect confidence intervals were compared with the short-horizon benchmarks quoted by Cochrane. This is a sensitivity analysis, not an endorsement of those thresholds.

Endpoint	Scenario	Benefit (95% CI), points	Threshold	Compatibility
ADAS-Cog scale at 18 months	Class pool	1.09 (0.66, 1.51)	2 points	Threshold excluded by 95% CI
ADAS-Cog scale at 18 months	Class pool	1.09 (0.66, 1.51)	3 points	Threshold excluded by 95% CI
ADAS-Cog scale at 18 months	Class pool	1.09 (0.66, 1.51)	4 points	Threshold excluded by 95% CI
ADAS-Cog scale at 18 months	Lecanemab + donanemab	1.40 (0.80, 1.99)	2 points	Threshold excluded by 95% CI
ADAS-Cog scale at 18 months	Lecanemab + donanemab	1.40 (0.80, 1.99)	3 points	Threshold excluded by 95% CI
ADAS-Cog scale at 18 months	Lecanemab + donanemab	1.40 (0.80, 1.99)	4 points	Threshold excluded by 95% CI
ADAS-Cog scale at 18 months	Response primary	1.31 (0.85, 1.77)	2 points	Threshold excluded by 95% CI
ADAS-Cog scale at 18 months	Response primary	1.31 (0.85, 1.77)	3 points	Threshold excluded by 95% CI
ADAS-Cog scale at 18 months	Response primary	1.31 (0.85, 1.77)	4 points	Threshold excluded by 95% CI
CDR-SB scale at 18 months	Class pool	0.29 (0.00, 0.58)	1 points	Threshold excluded by 95% CI
CDR-SB scale at 18 months	Class pool	0.29 (0.00, 0.58)	2 points	Threshold excluded by 95% CI
CDR-SB scale at 18 months	Lecanemab + donanemab	0.56 (0.32, 0.81)	1 points	Threshold excluded by 95% CI
CDR-SB scale at 18 months	Lecanemab + donanemab	0.56 (0.32, 0.81)	2 points	Threshold excluded by 95% CI
CDR-SB scale at 18 months	Response primary	0.39 (-0.09, 0.86)	1 points	Threshold excluded by 95% CI
CDR-SB scale at 18 months	Response primary	0.39 (-0.09, 0.86)	2 points	Threshold excluded by 95% CI

At 18 months, every evaluated ADAS-Cog CI was below the 2-point MCI benchmark; the lecanemab-donanemab upper bound was 1.99 points. Every CDR-SB CI was below the 1-point MCI benchmark. These comparisons are sensitive to model choice and to the mismatch between the benchmarks' cited 6-12-month derivation and the 18-month trial horizon.

For outcomes beyond 24 months, the CIs were sometimes compatible with the lower ADAS-Cog threshold, but only two studies contributed and the threshold was not validated for that horizon. Longer follow-up therefore does not solve the threshold problem in this dataset.

8. Biological target-class analyses

Endpoint	Target class	k	Estimate (95% CI)
ADAS-Cog scale at 18 months	Aggregated/fibrillar Aβ	8	-0.063 (-0.146, 0.020)
ADAS-Cog scale at 18 months	Pyroglutamate plaque Aβ	1	-0.175 (-0.284, -0.065)
ADAS-Cog scale at 18 months	Soluble aggregate-preferring	2	-0.166 (-0.260, -0.072)
ADAS-Cog scale at 18 months	Soluble monomer Aβ	2	-0.114 (-0.205, -0.024)
CDR-SB scale at 18 months	Aggregated/fibrillar Aβ	4	-0.039 (-0.321, 0.243)
CDR-SB scale at 18 months	Pyroglutamate plaque Aβ	1	-0.293 (-0.404, -0.182)
CDR-SB scale at 18 months	Soluble aggregate-preferring	2	-0.365 (-0.905, 0.174)
CDR-SB scale at 18 months	Soluble monomer Aβ	2	-0.075 (-0.141, -0.008)
Any ARIA E at 18 months	Aggregated/fibrillar Aβ	6	12.453 (7.754, 19.998)
Any ARIA E at 18 months	Pyroglutamate plaque Aβ	1	11.669 (7.275, 18.717)
Any ARIA E at 18 months	Soluble aggregate-preferring	2	7.337 (4.347, 12.383)
Any ARIA E at 18 months	Soluble monomer Aβ	2	2.007 (0.682, 5.912)
Study discontinuation at 18 months	Aggregated/fibrillar Aβ	6	1.104 (0.977, 1.248)
Study discontinuation at 18 months	Pyroglutamate plaque Aβ	1	1.360 (1.144, 1.617)
Study discontinuation at 18 months	Soluble aggregate-preferring	2	1.224 (1.018, 1.471)
Study discontinuation at 18 months	Soluble monomer Aβ	2	0.959 (0.825, 1.115)

Soluble aggregate-preferring and pyroglutamate-plaque categories showed favorable cognitive estimates, while the broad aggregated/fibrillar category was closer to null. However, this category combines aducanumab, bapineuzumab, and gantenerumab, whose dose exposure and plaque removal differ substantially. Solanezumab, classified as soluble-monomer targeting, also produced small favorable estimates on some scales. Target class therefore does not replace individual-drug analysis.

The classifications summarize predominant binding preferences and are not mutually exclusive biochemical truths. Several antibodies bind more than one amyloid species, and in-vivo effects depend on affinity, exposure, effector function, and disease context.

9. Moderator and functional-unblinding analyses

Endpoint	Moderator	k	Slope (95% CI)	P
ADAS-Cog scale at 18 months	Biomarker-confirmed enrollment (binary)	13	-0.0538 (-0.1598, 0.0521)	0.2871
ADAS-Cog scale at 18 months	Futility termination (binary)	13	-0.0041 (-0.1394, 0.1312)	0.9478
ADAS-Cog scale at 18 months	Trial publication year per 5 years	13	-0.0628 (-0.1251, -0.0004)	0.0487
ADAS-Cog scale at 18 months	ARIA-E risk difference per 10 percentage points	11	-0.0088 (-0.0588, 0.0411)	0.6983
MMSE scale at 18 months	Biomarker-confirmed enrollment (binary)	8	-0.0626 (-0.2545, 0.1292)	0.4547
MMSE scale at 18 months	Futility termination (binary)	8	-0.0631 (-0.2416, 0.1154)	0.4206
MMSE scale at 18 months	Trial publication year per 5 years	8	-0.0102 (-0.1715, 0.1511)	0.8817
MMSE scale at 18 months	ARIA-E risk difference per 10 percentage points	6	-0.0115 (-0.0709, 0.0479)	0.6192
CDR-SB scale at 18 months	Biomarker-confirmed enrollment (binary)	9	-0.0890 (-0.4514, 0.2734)	0.5797
CDR-SB scale at 18 months	Futility termination (binary)	9	0.0663 (-0.2039, 0.3365)	0.5799
CDR-SB scale at 18 months	Trial publication year per 5 years	9	-0.1310 (-0.3009, 0.0389)	0.1110
CDR-SB scale at 18 months	ARIA-E risk difference per 10 percentage points	7	-0.0081 (-0.1190, 0.1027)	0.8578
ADCS-ADL-MCI score at 18 months	Futility termination (binary)	4	-0.0545 (-0.4421, 0.3331)	0.6065
ADCS-ADL-MCI score at 18 months	Trial publication year per 5 years	4	0.2298 (-1.7676, 2.2271)	0.6697

The ARIA-E risk-difference proxy was not associated with ADAS-Cog, MMSE, or CDR-SB effects. This does not establish successful blinding: ARIA-E is an imperfect proxy, clinical teams may remain blinded to central MRI reads, infusion reactions may unblind independently, and trial-level meta-regression cannot recover participant-level expectancy effects.

Publication year showed a nominal association with larger ADAS-Cog effects (P=0.049), but this was one of multiple exploratory moderators and is inseparable from changes in biomarker selection, antibodies, dose exposure, and trial design. Biomarker-confirmation and futility indicators were not statistically significant moderators.

10. Influence, heterogeneity, and certainty

Leave-one-antibody-out results demonstrate which class estimates are composition-sensitive. They should not be read as independent replications because each estimate reuses most of the same trials.

Endpoint	Omitted antibody	k	Estimate (95% CI)	I2
ADAS-Cog scale at 18 months	Aducanumab	10	-0.105 (-0.161, -0.049)	16.8
ADAS-Cog scale at 18 months	Bapineuzumab	9	-0.133 (-0.183, -0.084)	4.1
ADAS-Cog scale at 18 months	Crenezumab	12	-0.110 (-0.161, -0.059)	11.4
ADAS-Cog scale at 18 months	Donanemab	12	-0.101 (-0.154, -0.048)	10.6
ADAS-Cog scale at 18 months	Gantenerumab	12	-0.114 (-0.165, -0.063)	10.6
ADAS-Cog scale at 18 months	Lecanemab	12	-0.100 (-0.154, -0.046)	10.3
ADAS-Cog scale at 18 months	Solanezumab	11	-0.106 (-0.171, -0.041)	17.8
CDR-SB scale at 18 months	Aducanumab	6	-0.139 (-0.300, 0.022)	73.9
CDR-SB scale at 18 months	Crenezumab	8	-0.116 (-0.230, -0.002)	69.5
CDR-SB scale at 18 months	Donanemab	8	-0.096 (-0.202, 0.011)	44.4
CDR-SB scale at 18 months	Gantenerumab	8	-0.141 (-0.266, -0.017)	66.1
CDR-SB scale at 18 months	Lecanemab	8	-0.110 (-0.254, 0.035)	68.0
CDR-SB scale at 18 months	Solanezumab	7	-0.146 (-0.332, 0.041)	69.4
Any ARIA H at 18 months	Crenezumab	2	2.138 (1.784, 2.563)	25.6
Any ARIA H at 18 months	Donanemab	2	1.330 (0.603, 2.938)	84.1
Any ARIA H at 18 months	Lecanemab	2	1.455 (0.547, 3.869)	90.2

The response's criticism of moderate certainty for a highly inconsistent ARIA-H estimate is not itself a statistical hypothesis. The dataset can document the inconsistency, prediction intervals, and influence of individual agents; a revised GRADE rating additionally requires judgments about risk of bias, indirectness, imprecision, and the decision context.

For class-wide ARIA-H at 18 months, I2 was 90%. The estimate became more coherent only after the analysis effectively narrowed to lecanemab and donanemab. This supports reporting treatment-specific absolute risks and argues against a single class safety label.

11. What the Cochrane dataset cannot answer

Question	Why it remains unresolved
Individual-level amyloid mediation	Trial-level Centiloid associations cannot show that participants with greater clearance had greater benefit.
Continuous amyloid-positive enrollment fraction	The export records whether confirmation was required, not a consistent participant-level fraction for every older trial.
Personalized safety	APOE genotype, anticoagulation, baseline microhemorrhage burden, MRI surveillance, and event severity are unavailable.
Patient-valued net benefit	No defensible universal weights exist for cognitive slowing, ARIA, infusion burden, cost, and access.
Real-world effectiveness	Randomized trial summaries cannot substitute for ALZ-NET or comparable post-approval observational data.
Guideline recommendation	Evidence synthesis alone does not reproduce multidisciplinary guideline methods or stakeholder deliberation.

These are not merely missing statistical techniques. They require different data, explicit value judgments, or a different methodological process.

12. Conclusions

The expanded analyses reinforce the need to distinguish individual anti-amyloid antibodies, their biological activity, and their absolute harms. Lecanemab and donanemab individually showed favorable estimates on central cognitive and global endpoints, but the dataset was too small to establish statistically significant treatment-by-antibody interactions. Excluding futility trials did not materially strengthen ADAS-Cog and only modestly changed CDR-SB, suggesting that biological target engagement is a more informative distinction than termination status alone.

Absolute safety estimates revealed clinically important differences: ARIA-E excess risk was approximately 109 per 1,000 for lecanemab and 220 per 1,000 for donanemab, with an even larger estimate for aducanumab. The raw efficacy confidence intervals remained below the short-horizon ADAS-Cog and CDR-SB benchmarks cited by Cochrane at 18 months, although the response is correct that threshold selection and time-horizon transfer require external validation and stakeholder judgment.

Overall, the data justify a treatment-specific, biologically informed systematic review. They do not justify a universal conclusion that all anti-amyloid antibodies are interchangeable, nor do they establish an unqualified favorable benefit-risk balance for the active agents.

Sources

Source	Use	URL
Cochrane CD016297 (2026)	Review, row-level data, trial termination, MID discussion	https://doi.org/10.1002/14651858.CD016297
Snyder et al. (2026)	Criticisms and requested alternative framing	https://doi.org/10.1002/alz.71696
Cummings et al. (2025)	Predominant amyloid-species target classification	https://pmc.ncbi.nlm.nih.gov/articles/PMC12047478/
Muir et al. (2024)	MID/MCID source cited by Cochrane	https://doi.org/10.1002/alz.13770
Lansdall et al. (2023)	Clinically meaningful change source cited by Cochrane	https://doi.org/10.14283/jpad.2022.102
Imbimbo et al. (2024)	Trial-level Centiloid mapping from prior package	https://academic.oup.com/brain/article/147/10/3513/7754406

Reproducibility

The companion archive contains the annotated trial map, every result table, analysis code, workbook builder, report builder, and figures. Quality-control checks confirmed 17 classified studies, seven antibodies, six futility terminations, nine total early terminations, and no missing agent or termination classifications.